

EcoPassEU: A Privacy-Aware Digital Product Passport DApp for Sustainable Consumer Goods in Europe

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1 Introduction

In this section, introduce the DApp and explain its importance without being overly technical. Focus on its uses and applications, addressing what EcoPassEU is, why combating greenwashing is important, and identifying the target user groups (e.g., European manufacturers, SMEs, consumers, and regulators).

2 Background

Describe the landscape of sustainable consumer goods and the Ecodesign for Sustainable Products Regulation (ESPR). Detail the necessity of DPPs, present similar traceability applications or research, and discuss the business and financial viability of the DApp as a potential startup. Remember to include citations.

3 Design Documentation

Outline the system architecture of EcoPassEU. Include your justifications for using EVM-compatible networks like Polygon, ERC-1155 tokens, IPFS, and DIDs/VCs. Insert your Use Case diagrams, Sequence/Flow diagrams, and front-end wireframes here.

4 Project Execution

Detail the implementation process of the DApp. This should feature core code analysis, data structures, your testing strategy, final UI screenshots, and a brief user manual on how to interact with the application.

4.1 Critical Evaluation

Provide a technical analysis of EcoPassEU, evaluating the gas usage of your smart contracts, security considerations (especially regarding the separation of on-chain and off-chain data), and runtime performance.

5 Conclusion

Summarize the main theoretical and technical findings. Detail what the group achieved, acknowledge limitations (e.g., the physical-to-digital bridge vulnerabilities), and propose future work.

References

1. European Parliament and Council: Regulation (EU) 2024/1781 establishing a framework for the setting of ecodesign requirements for sustainable products. Official Journal of the European Union (2024).
2. European Commission: Implementing the Ecodesign for Sustainable Products Regulation. European Commission Green Forum (2026).
3. European Commission: The EU Digital Identity Framework Regulation Enters into Force. EU Digital Identity Wallet (2024).
4. European Commission: European Blockchain Services Infrastructure (EBSI) Homepage. (2026).
5. W3C: Decentralized Identifiers (DIDs) v1.0. W3C Recommendation (2022).
6. W3C: Verifiable Credentials Data Model v2.0. W3C Recommendation (2025).

A Appendix

This appendix includes the mandatory materials required for the FTGP summative assessment.

A.1 Smart Contract Deployment

- **Smart Contract Address(es):** [Insert Sepolia testnet addresses here]
- **Etherscan Links:** [Insert links to verified contracts on Etherscan]

A.2 Project Links

- **Deployed Website:** [Insert URL]
- **GitHub Repository:** https://github.com/DeTaiDong/FTGP2425_Group1_11

A.3 Transaction Hashes

- **Deployment TX:** [Insert Hash] - Explanation.

A.4 Equity Shares & Contributions

- Luxiao Cao: [Weight]
- Akshansh Rajora: [Weight]
- Fuyu Cao: [Weight]
- Tianwei Yu: [Weight]
- Detai Dong: [Weight]

[Insert Screenshot of GitHub - Insights - Contributor here]

A.5 JIRA Project Board

[Insert screenshots of 'Summary', 'timeline', and 'backlog' pages here]

A.6 Individual Reflection Essays

Luxiao Cao [Insert 500-word reflection here]

Akshansh Rajora [Insert 500-word reflection here]

Fuyu Cao [Insert 500-word reflection here]

Tianwei Yu [Insert 500-word reflection here]

Detai Dong [Insert 500-word reflection here]

A.7 Team Sprint Reports

[Insert chronological sprint reports here]

A.8 Complete Source Code

Smart Contracts

```
// Insert complete Solidity code here
```

Frontend Core Files

```
// Insert modified/relevant frontend files (e.g., page.js) here
```